LeafNet: Lightweight Pruning of Convolutional Networks for Edge Devices

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Abstract

We propose LeafNet, a pruning method that reduces the size of convolutional networks by 62 percent while losing less than 1 percent accuracy on image classification. The method targets low-power edge devices such as smart cameras.

1 Introduction

Edge devices such as smart cameras and drones usually have less than 2 GB of memory and a weak processor. Large convolutional networks are difficult to run on them.

Pruning removes parts of a network that contribute little to its output. Many pruning methods are complex and need long retraining. In this paper we ask whether a simple and cheap pruning rule can work well.

2 Method

LeafNet ranks every filter by its mean activation on a small calibration set and removes the lowest 40 percent of filters in each layer.

After pruning, the network is fine-tuned for 15 epochs using a learning rate of 0.001 with the Adam optimizer. No other changes to the architecture are made.

3 Results

On the CIFAR-10 dataset, the pruned ResNet-18 reaches 93.4 percent accuracy compared with 94.1 percent for the original model.

Inference time on a Raspberry Pi 4 drops from 210 ms to 78 ms per image, and the model file becomes 2.6 times smaller.

4 Conclusion

LeafNet shows that simple activation-based pruning is enough to make convolutional networks small and fast with a very small loss of accuracy.

Future work will study pruning of transformer models and the use of larger datasets.

References

[1] S. Han et al. Learning both weights and connections for efficient neural networks. 2015.

[2] K. He et al. Deep residual learning for image recognition. 2016.